

Problems

(S.P 3.02, 3.03, 3.04, 3.05, 3.07)
1, 2, 3, 6, 8, 9, 10, 11, 12, 13, 14, 15,
16, 34, 35, 36, 37, 38, 39)

1 - Given

$$\theta = 250^\circ$$

$$|a| = 7.3 \text{ cm}$$

$$a_x = a_y = ?$$

Sol

$$a_x = a \cos \theta$$

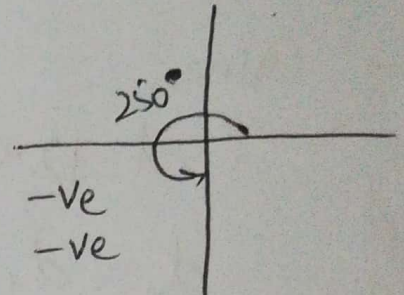
$$= 7.3 \cos 250^\circ$$

$$\boxed{a_x = -2.5 \text{ m}}$$

$$, a_y = a \sin \theta$$

$$= 7.3 \sin 250^\circ$$

$$, \boxed{a_y = -6.9 \text{ m}}$$



2 - Given

$$r = 15 \text{ m}, \theta = 30^\circ$$

$$a_x = a_y = ?$$

Sol

If $\vec{r}$ is given

$$r_x = r \cos \theta$$

$$= 15 \cos 30^\circ$$

$$\boxed{r_x = 13 \text{ m}}$$

$$, r_y = r \sin \theta$$

$$= 15 \sin 30^\circ$$

$$, \boxed{r_y = 7.5 \text{ m}}$$

3 - Given

$$A_x = -25 \text{ m}, A_y = 40 \text{ m}$$

$$|A| = ?$$

$$\theta = ?$$

$$|A| = \sqrt{A_x^2 + A_y^2}$$

$$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$$

b- Given

$$d = 12.5 \text{ m}$$

$$\theta = 20.0^\circ$$

$$d_x = d_y = ?$$

Sol

$$d_x = d \cos \theta$$

$$, d_y = d \sin \theta$$

c- Given

$$A_x =$$

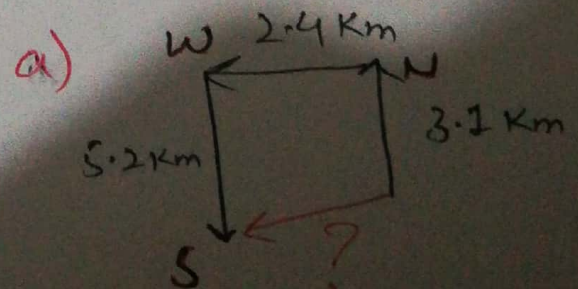
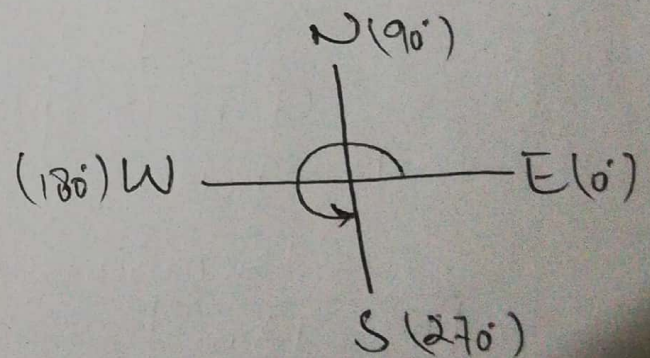
$$A_y =$$

$$A_z =$$

Sol

$$\rightarrow |A| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

$$\rightarrow \theta = 42^\circ \text{ (South of West)}$$



Given

$$\vec{a} = (4\text{m})\hat{i} - (3\text{m})\hat{j} + (1\text{m})\hat{k}$$

$$\vec{b} = -1\hat{i} + 1\hat{j} + 4\hat{k}$$

Sol

$$a) \vec{a} + \vec{b} = 3\hat{i} - 2\hat{j} + 5\hat{k}$$

$$b) \vec{a} - \vec{b} = 5\hat{i} - 4\hat{j} - 3\hat{k}$$

$$c) \vec{c} = \vec{b} - \vec{a} \Rightarrow -5\hat{i} + 4\hat{j} + 3\hat{k}$$

10- $r_x = r_y = r_z = ?$

$$\vec{c} = c_x + c_y + c_z = 7 \cdot 4\hat{i} - 3 \cdot 8\hat{j} - 6 \cdot 1\hat{k}$$

$$\vec{d} = 4 \cdot 4\hat{i} - 2\hat{j} + 3 \cdot 3\hat{k}$$

Sol

$$r_x = c_x + d_x = 11.8\text{m}$$

$$r_y = c_y + d_y = -5.8\text{m}$$

$$r_z = c_z + d_z = -2.8\text{m}$$

11- Given

$$a = 4\hat{i} + 3\hat{j}$$

$$b = -13\hat{i} + 7\hat{j}$$

$$\vec{a} + \vec{b} = ? , |\vec{a} + \vec{b}| = ?$$

$$\theta =$$

Sol

$$r_x = a_x + b_x = -9, \quad r_y = a_y + b_y = 10$$

$$|r| = \sqrt{r_x^2 + r_y^2}$$

$$\theta = \tan^{-1}\left(\frac{r_y}{r_x}\right)$$

$$= \sqrt{(-9)^2 + (10)^2}$$

$$\theta = \tan^{-1}\left(\frac{10}{-9}\right)$$

$$|r| = 13.45$$

$$= -48^\circ \text{ or } 132^\circ$$

It make sense
because of
Quadrant

H.W

12- Given (Similar to Ex. 8)

$$A = (150 \text{ Km})\hat{i}$$

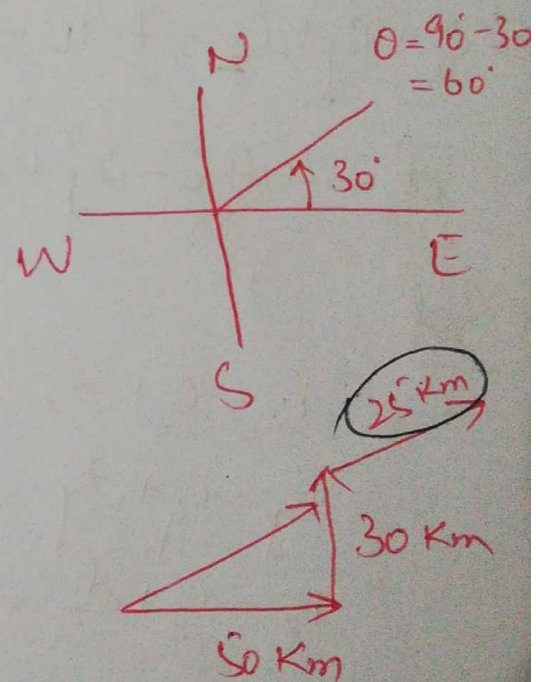
$$B = (30 \text{ Km})\hat{j}$$

$$C = ?$$

Sol

$$C = C_x\hat{i} + C_y\hat{j}$$

$$C = 25 \cos 60^\circ \hat{i} + 25 \sin 60^\circ \hat{j}$$



Given

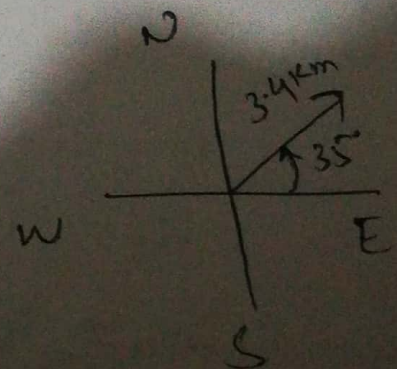
13

$$d = 3.4 \text{ km}, \quad \theta = 35^\circ$$

$$\text{Sol} \rightarrow d = d \cos \theta + d \sin \theta$$

$$= 3.4 \cos 35^\circ + 3.4 \sin 35^\circ$$

$$= 4.74 \text{ km}$$



Given

$$(-140, 30)$$

Total x distance

Total y distance

$$(20, 60), (b_x, -70), (-20, c_y), (-60, -70)$$

Sol.

$$a) \text{ Total 'x' distance} = 20 + b_x - 20 - 60 = 140$$

$$b_x = -140 + 60$$

$$\boxed{b_x = -80 \text{ m}}$$

$$b) \text{ Total 'y' distance} = 60 - 70 + c_y - 70 = 30$$

$$c_y = 30 - 60 + 140$$

$$\boxed{c_y = 110 \text{ m}}$$

$$c) \sqrt{(-140)^2 + (30)^2} \Rightarrow \sqrt{19600 + 900}$$
$$= \sqrt{20500}$$
$$= 143.18 \text{ m}$$

$$d) \theta = \tan^{-1}\left(\frac{30}{-140}\right) = -12^\circ + 180^\circ \Rightarrow \boxed{168^\circ}$$

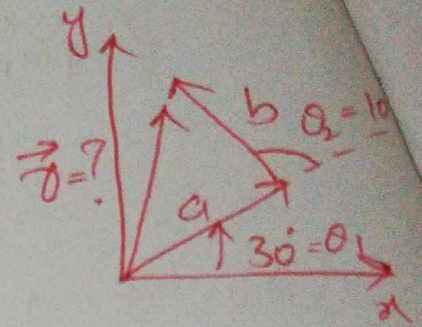
15 Magnitude = 10 m, $\theta_1 = 30^\circ$, $\theta_2 = 105^\circ$

$$a = a_x \hat{i} + a_y \hat{j}$$

$$= a \cos \theta \hat{i} + a \sin \theta \hat{j}$$

$$b = b_x \hat{i} + b_y \hat{j}$$

$$= b \cos \theta \hat{i} + b \sin \theta \hat{j}$$



Sol

$$= \{a \cos \theta_1 + b \cos (\theta_1 + \theta_2)\} \hat{i} + \{a \sin \theta_1 + b \sin (\theta_1 + \theta_2)\} \hat{j}$$

$$= \{10 \cos 30^\circ + 10 \cos (30^\circ + 105^\circ)\} \hat{i} + \{10 \sin 30^\circ + 10 \sin (30^\circ + 105^\circ)\} \hat{j}$$

$$= \{1.54 + (-9.9)\} \hat{i} + \{-9.88 + 0.88\} \hat{j}$$

CH#3 (Problems)

$$\vec{a} = 3\hat{i} + 5\hat{j}$$

$$\vec{b} = 2\hat{i} + 4\hat{j}$$

$$\begin{aligned} \text{a) } \vec{a} \times \vec{b} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 5 & 0 \\ 2 & 4 & 0 \end{vmatrix} \\ &= (12 - 10)\hat{k} \Rightarrow 2\hat{k} \end{aligned}$$

$$\begin{aligned} \text{b) } \vec{a} \cdot \vec{b} &= 6 + 20 \\ &= 26 \end{aligned}$$

$$\begin{aligned} \text{c) } (\vec{a} + \vec{b}) \cdot \vec{b} &= (5\hat{i} + 9\hat{j}) \cdot (2\hat{i} + 4\hat{j}) \\ &= 10 + 36 \\ &= 46 \end{aligned}$$

$$\text{d) } a \cos \theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \Rightarrow \frac{26}{\sqrt{20}}$$

$$35- \quad |\vec{r}| = 4.5 \text{ m}$$

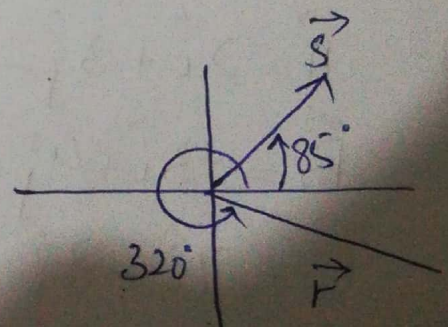
$$|\vec{s}| = 7.3 \text{ m}$$

$$\vec{r} \cdot \vec{s} = rs \cos \theta$$

$$= (4.5)(7.3) \cos 235^\circ \Rightarrow$$

$$\vec{r} \times \vec{s} = rs \sin \theta$$

$$= (4.5)(7.3) \sin 235^\circ \Rightarrow$$



Angle b/w $\vec{r}$ & $\vec{s}$

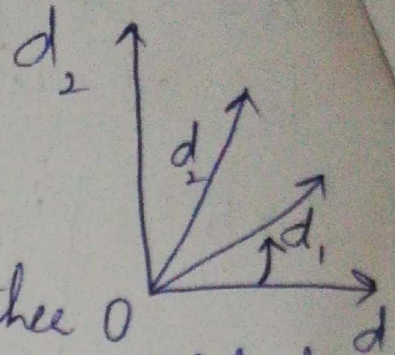
$$320 - 85 \Rightarrow$$

or

$$360 - 320 = 40 + 85 = 125^\circ$$

$$36- \vec{d}_1 = 3\hat{i} - 2\hat{j} + 4\hat{k}, \vec{d}_2 = -5\hat{i} + 2\hat{j} - \hat{k}$$

$$(\vec{d}_1 + \vec{d}_2) \cdot (\vec{d}_1 \times 4\vec{d}_2) = ?$$



The answer is zero because when d_1 & d_2 multiply with each other 0 in xy -plane give $\hat{k}$ and the resultant of $d_1 + d_2$ is also in xy -plane so it make 90° degree angle with $\hat{k}$ so $\cos 90^\circ = 0$

$$37- \vec{a} = 3\hat{i} + 3\hat{j} - 2\hat{k}$$

$$\vec{b} = -2\hat{i} - 4\hat{j} + 2\hat{k}$$

$$\vec{c} = 2\hat{i} + 2\hat{j} + \hat{k}$$

$$a) \vec{a} \cdot (\vec{b} \times \vec{c}) =$$

$$b) \vec{a} \cdot (\vec{b} + \vec{c}) =$$

$$c) \vec{a} \times (\vec{b} + \vec{c}) =$$

$$38- \vec{A} = 2\hat{i} + 3\hat{j} - 4\hat{k}$$

$$\vec{B} = -3\hat{i} + 4\hat{j} + 2\hat{k}$$

$$\vec{C} = 7\hat{i} - 8\hat{j}$$

$$3\vec{C} \cdot (2\vec{A} \times \vec{B}) = ?$$